# Production Layer Decomposition

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# PHYS5033

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# Recap – Week 8

Footprint applications:

- Sustainability reporting
- GHG Protocol (Scope 1, 2 & 3)
- TCFD
- TNFD
- Real life example

Starting to think about how to calculate your personal footprint

# Footprint calculations

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1} y$$

$$\mathbf{x} = \mathbf{L} y$$

Multiply both sides by **q**

$$\mathbf{q}\mathbf{x} = \mathbf{q}\mathbf{L} y$$

$$\mathbf{q}\mathbf{x} = \mathbf{m} y$$

$$Footprint = \mathbf{m} y = \mathbf{q}\mathbf{L} y$$

# Analytical Techniques

Understanding the mathematical features of **L** and **A** unlocks two additional analytical techniques within input-output analysis:

1) **Production Layer Decomposition (PLD)**
   - Provides the breakdown of total impacts according to production layers
   - Helps us understand how far up the supply chain the sectors that contribute the most to the overall footprint are

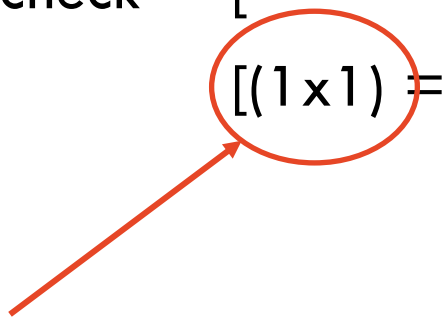
2) **Structural Path Analysis (SPA)**
   - Allows us to quantify the impacts embodied in specific supply chains

# Footprint calculation

$$\boldsymbol{Footprint = \mathbf{qL}y}$$

Size check [ = (1x6)*(6x6)*(6x1)]

[(1x1) = (1x6)*(6x6)*(6x1)]



This doesn't provide us with much detail!

# Taking a closer look at L

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

$$= \frac{1}{\mathbf{I} - \mathbf{A}}$$

Are there other ways we can express this which allow us to extract more information regarding the footprint?

# Let's go on a short maths adventure



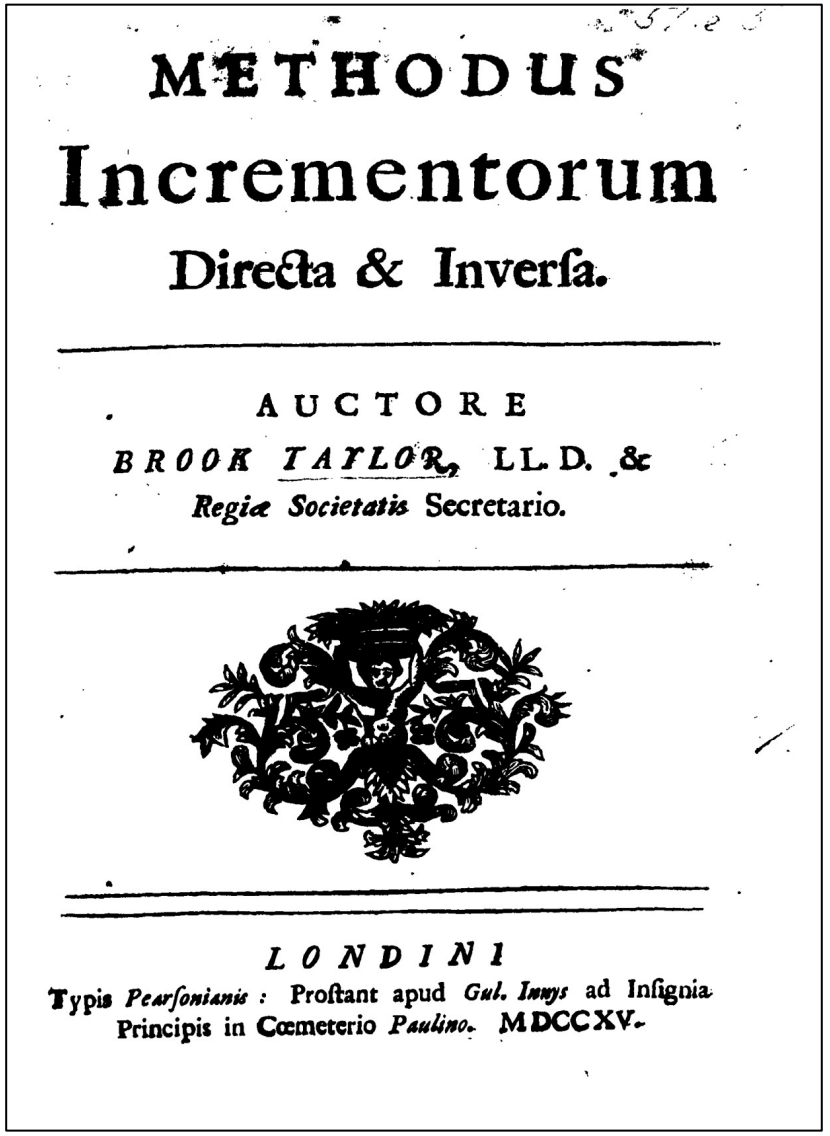
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# The Taylor Polynomial

Brook Taylor, 1685 - 1731

'Father' of the mathematical field of Calculus of finite differences. Key contributions include:

- Integration by parts
- Taylor polynomial
- Taylor series expansion



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Principis in Cœmeterio Paulino. MDCCXV.

# The Taylor Polynomial

$$T_{f,a}(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \cdots + \frac{1}{n!}f^n(a)(x-a)^n$$

Taylor presented this formula as a means of approximating the solution to complex functions, such as $f(a) = \frac{1}{1-a}$ , when $x$ is reasonably close to $a$.

Key concepts we need to understand:

- What does $f(a)$ represent?
- What does $f'(a)$ represent?
- What does $f^n(a)$ represent?
- What does $n!$ represent?

# The Taylor Polynomial

What does $f(a)$ represent?

In mathematics, $f(a)$ is a shorthand way of saying that something is a function of the variable $a$. For example,

$$y = a^2$$

can also be written as:

$$f(a) = a^2$$

# The Taylor Polynomial

What does $f'(a)$ represent?

In this context $f'(a)$ represents the first derivative of the function $f(a)$. The derivative is calculated following the rules of differentiation, the simplest of which is:

$$f'(x^n) = n \cdot (x^{n-1})$$

For example:

$$\begin{aligned} f'(x^3) &= 3 \cdot (x^{3-1}) \\ &= 3x^2 \end{aligned}$$

# What does a derivative tell us?

In simple terms the derivative tells us the rate of change of one variable with respect to another variable.

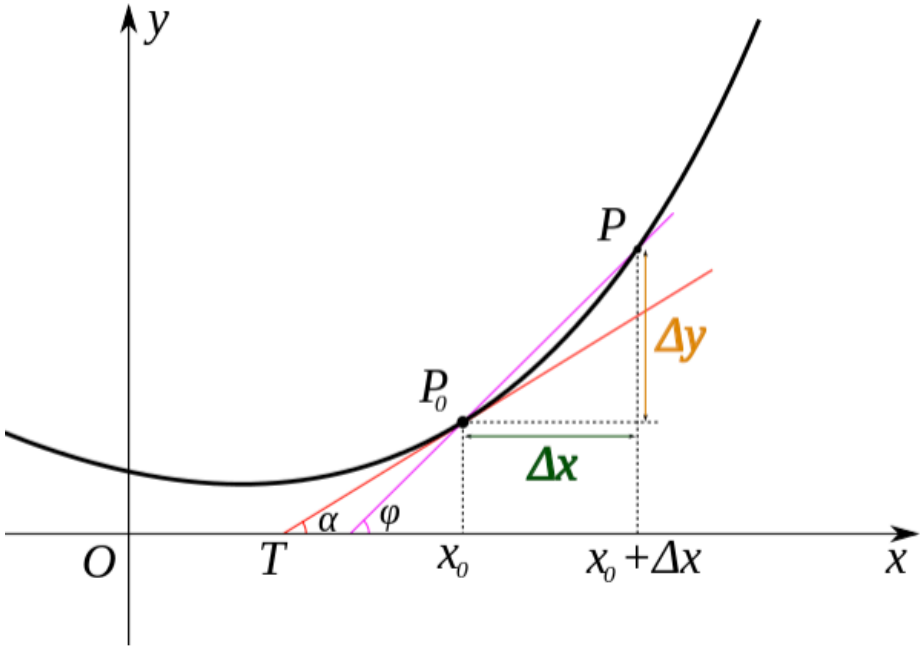
The rate of change of $y$ with respect to $(wrt)\, x$ can be calculated using

$$\frac{\Delta y}{\Delta x}$$

also written as

$$\frac{dy}{dx}$$

and $f'(y)$




(the $wrt\ x$ is assumed)

# The Taylor Polynomial

What does $f''(a)$ represent?

In this context $f''(a)$ represents the second derivative of the function $f(a)$. In the case of our example:

$$\begin{aligned} f''(x^3) &= f'(3x^2) \\ &= 3 \cdot 2 \cdot (x^{2-1}) \\ &= 6x^1 \\ &= 6x \end{aligned}$$

# The Taylor Polynomial

What does $f^n(a)$ represent?

In this context $f^n(a)$ represents the $n^{th}$ derivative of the function $f(a)$. In the case of our example, we would only need to calculate the third derivative before we would exhaust the calculation:

$$\begin{aligned} f'''(x^3) &= f'(6x^1) \\ &= 6 \cdot 1 \cdot (x^{1-1}) \\ &= 6x^0 \\ &= 6 \end{aligned}$$

# The Taylor Polynomial

What does $n!$ represent?

This notation represents the factorial of the number $n$, which is calculated by:

$$n! = n \times (n-1) \times (n-2) \times (n-3) \ \ldots \ 3 \times 2 \times 1$$

For example:

$$\begin{aligned} 6! &= 6 \times (6-1) \times (6-2) \times (6-3) \times (6-4) \times (6-5) \\ &= 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 720 \end{aligned}$$

# The Taylor Polynomial

$$T_{f,a}(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \cdots + \frac{1}{n!}f^n(a)(x-a)^n$$

Let's see if this works for our function:

$$f(x) = \frac{1}{1-x}$$

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{\mathbf{I}-\mathbf{A}}$$

If we set $a = 0$ in the Taylor polynomial this will give us:

$$T_{f,0}(x) = f(0) + f'(0)(x-0) + \tfrac{1}{2!}f''(0)(x-0)^2 + \tfrac{1}{3!}f'''(0)(x-0)^3 + \cdots$$

$$= f(0) + f'(0)(x) + \tfrac{1}{2!}f''(0)(x)^2 + \tfrac{1}{3!}f'''(0)(x)^3 + \cdots$$

# The Taylor Polynomial

$$T_{f,0}(x) = f(0) + f'(0)(x) + \frac{1}{2!}f''(0)(x)^2 + \frac{1}{3!}f'''(0)(x)^3 + \cdots$$

So, we need to calculate the following terms

$$f(x) = \frac{1}{1-x}$$

$$f(0) = \frac{1}{1-0}$$

$$= 1$$

$$f'(x) = \frac{1}{(1-x)^2} \qquad f'(0) = \frac{1}{(1-0)^2} \qquad = 1$$

$$f''(x) = 2 * \frac{1}{(1-x)^3} \qquad f''(0) = 2 * \frac{1}{(1-0)^3} \qquad = 2$$

$$f'''(x) = 2 * 3 * \frac{1}{(1-x)^4} \qquad f'''(0) = 2 * 3 * \frac{1}{(1-0)^4} \qquad = 6$$

# The Taylor Polynomial

Which gives us:

$$f(x) = \frac{1}{1-x}$$

$$T_{f,0}(x) = f(0) + f'(0)(x) + \frac{1}{2!}f''(0)(x)^2 + \frac{1}{3!}f'''(0)(x)^3 + \cdots$$

(1) (1) (2) (6)

$$= 1 + (1 * x) + (\frac{2}{2!}x^2) + (\frac{3*2}{3!}x^3) + (\frac{4*3*2}{4!}x^4) + \ldots$$

$$= 1 + x + x^2 + x^3 + x^4 + \ldots$$

$$= \sum_{i=0}^{N} x^i$$

# The Taylor Series Expansion

$$f(x) = \frac{1}{1-x}$$

$$= 1 + x + x^2 + x^3 + x^4 + \ldots$$

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

$$= \frac{1}{\mathbf{I}-\mathbf{A}}$$

$$= \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 + \ldots + \mathbf{A}^n$$

# End of short maths adventure



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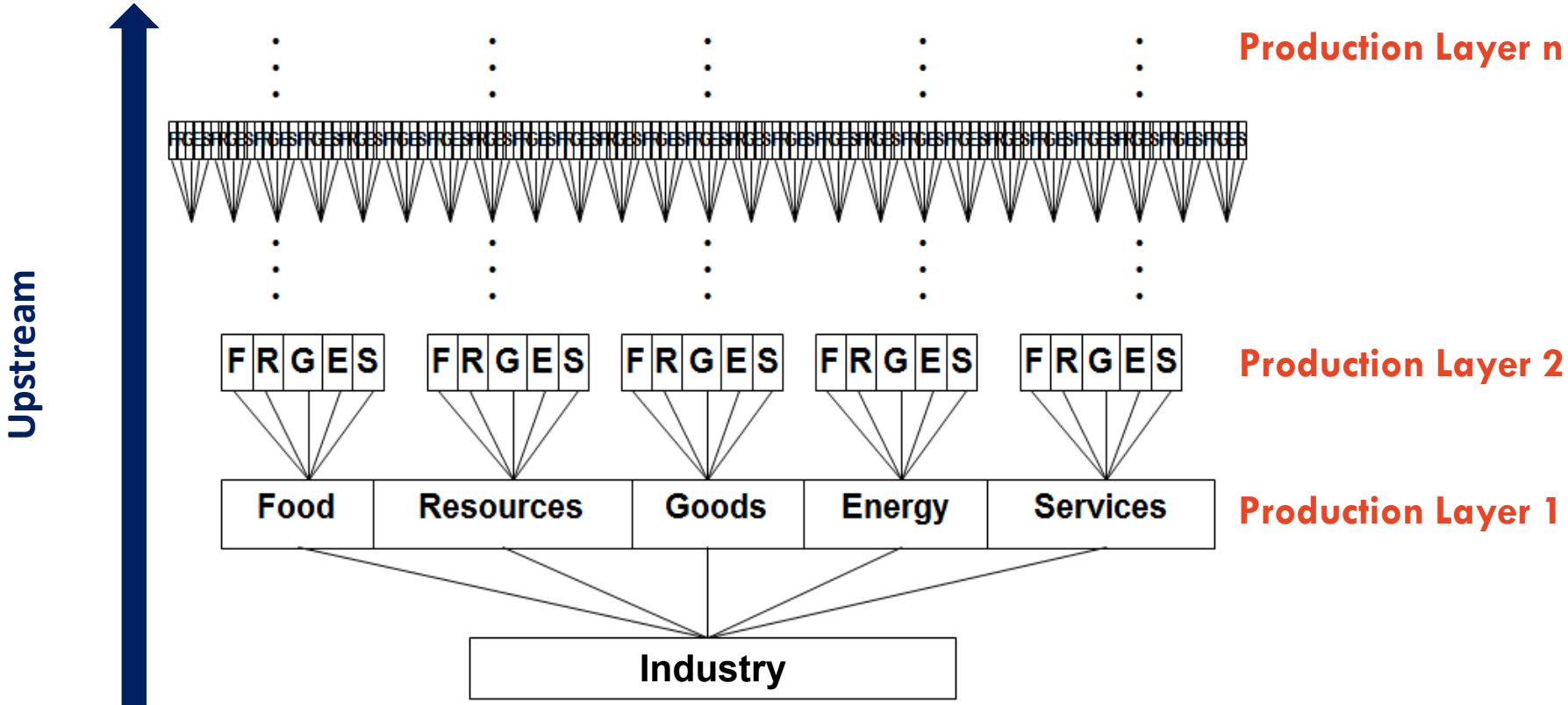
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Understanding the mathematical features of **L** and **A** unlocks two additional analytical techniques within input-output analysis:

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   - Helps us understand how far up the supply chain the sectors that contribute the most to the overall footprint are

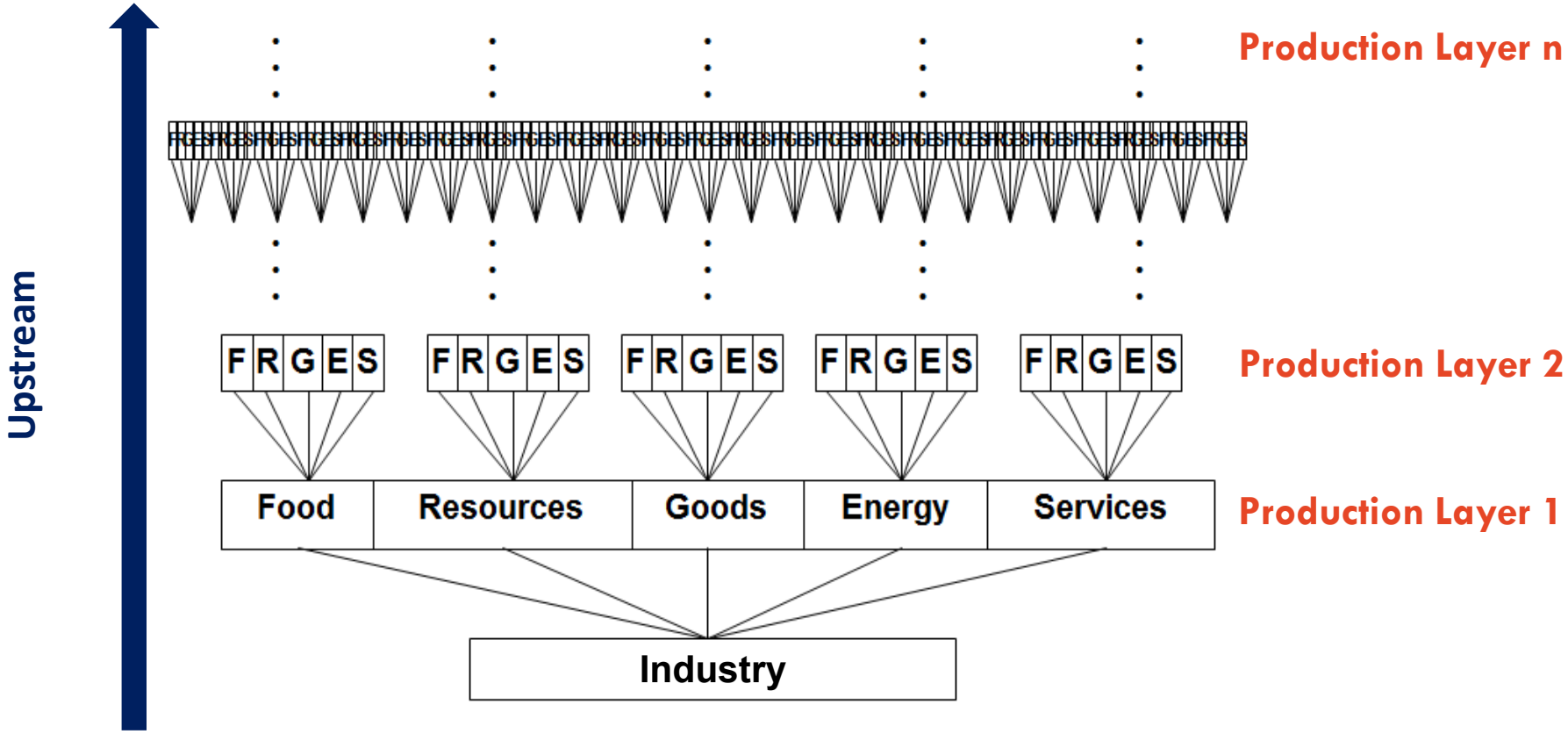
2) **Structural Path Analysis (SPA)**
   - Allows us to quantify the impacts embodied in specific supply chains

# Production Layer Decomposition




$$\mathbf{qLy}$$

# Production Layer Decomposition




$$\mathbf{qLy}$$

Layer 1 – 5 sectors
Layer 2 – 25 sectors
Layer 3 – 125 sectors
…

# Decomposing the footprint calculation

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

$$= \frac{1}{\mathbf{I}-\mathbf{A}}$$

$$= \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 + \ldots + \mathbf{A}^n$$

$\mathbf{qL}y$

Taylor series expansion of L

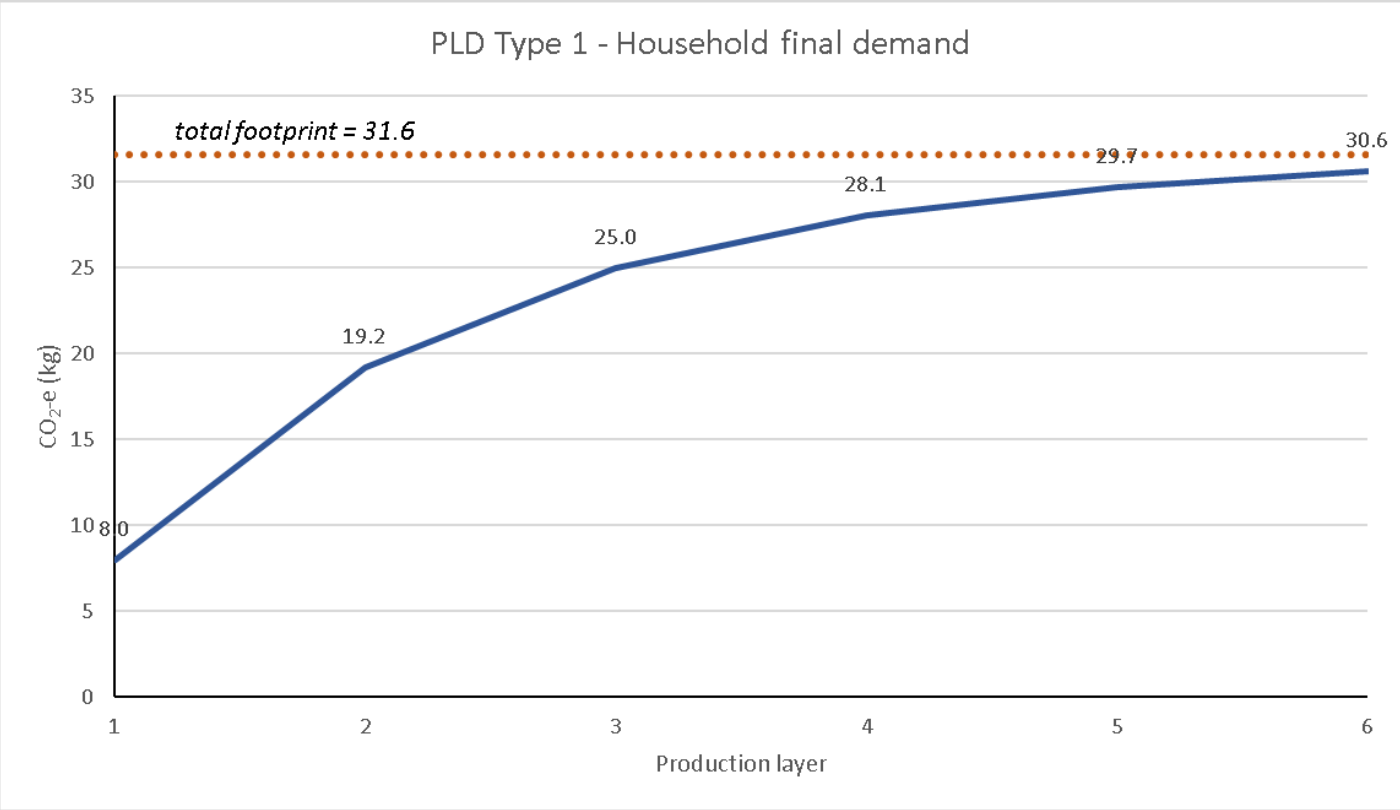
$$= \mathbf{q}(\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 + \cdots + \mathbf{A}^n)\boldsymbol{y}$$

$$= \mathbf{q}\mathbf{I}\boldsymbol{y} + \mathbf{q}\mathbf{A}\boldsymbol{y} + \mathbf{q}\mathbf{A}^2\boldsymbol{y} + \mathbf{q}\mathbf{A}^3\boldsymbol{y} + \mathbf{q}\mathbf{A}^4\boldsymbol{y} + \cdots + \mathbf{q}\mathbf{A}^n\boldsymbol{y}$$

# Decomposing the footprint calculation

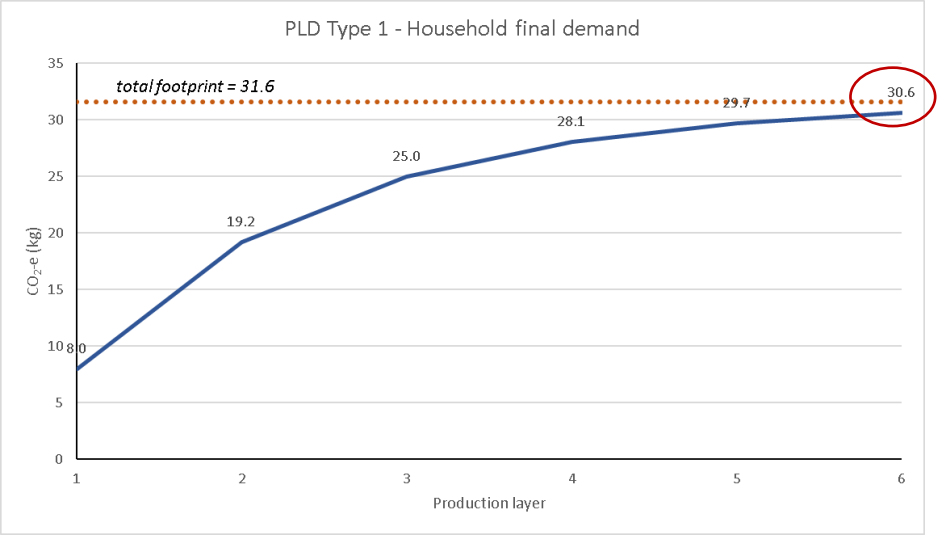
$$\mathbf{q}y + \mathbf{q}\mathbf{A}y + \mathbf{q}\mathbf{A}^2y + \mathbf{q}\mathbf{A}^3y + \mathbf{q}\mathbf{A}^4y + \cdots + \mathbf{q}\mathbf{A}^ny$$

Layer 1 Layer 2 Layer 3 Layer 4 Layer 5 Layer *n+1*



# Decomposing the footprint calculation

$$\underbrace{\mathbf{q}y}_{\text{Layer 1}} + \underbrace{\mathbf{q}\mathbf{A}y}_{\text{Layer 2}} + \underbrace{\mathbf{q}\mathbf{A}^2y}_{\text{Layer 3}} + \underbrace{\mathbf{q}\mathbf{A}^3y}_{\text{Layer 4}} + \underbrace{\mathbf{q}\mathbf{A}^4y}_{\text{Layer 5}} + \cdots + \underbrace{\mathbf{q}\mathbf{A}^ny}_{\text{Layer } n+1}$$




Very close to the total footprint !

**Plot cumulative results**

**Why does the graph converge?**

# Differential Convergence of Life-Cycle Inventories toward Upstream Production Layers

## Implications for Life-Cycle Assessment

*Manfred Lenzen and Graham Treloar*

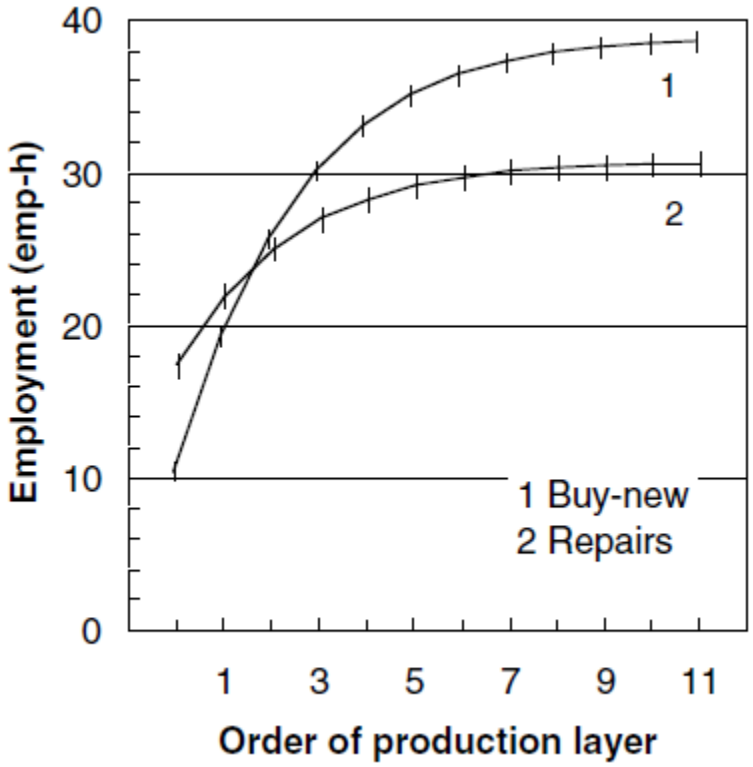


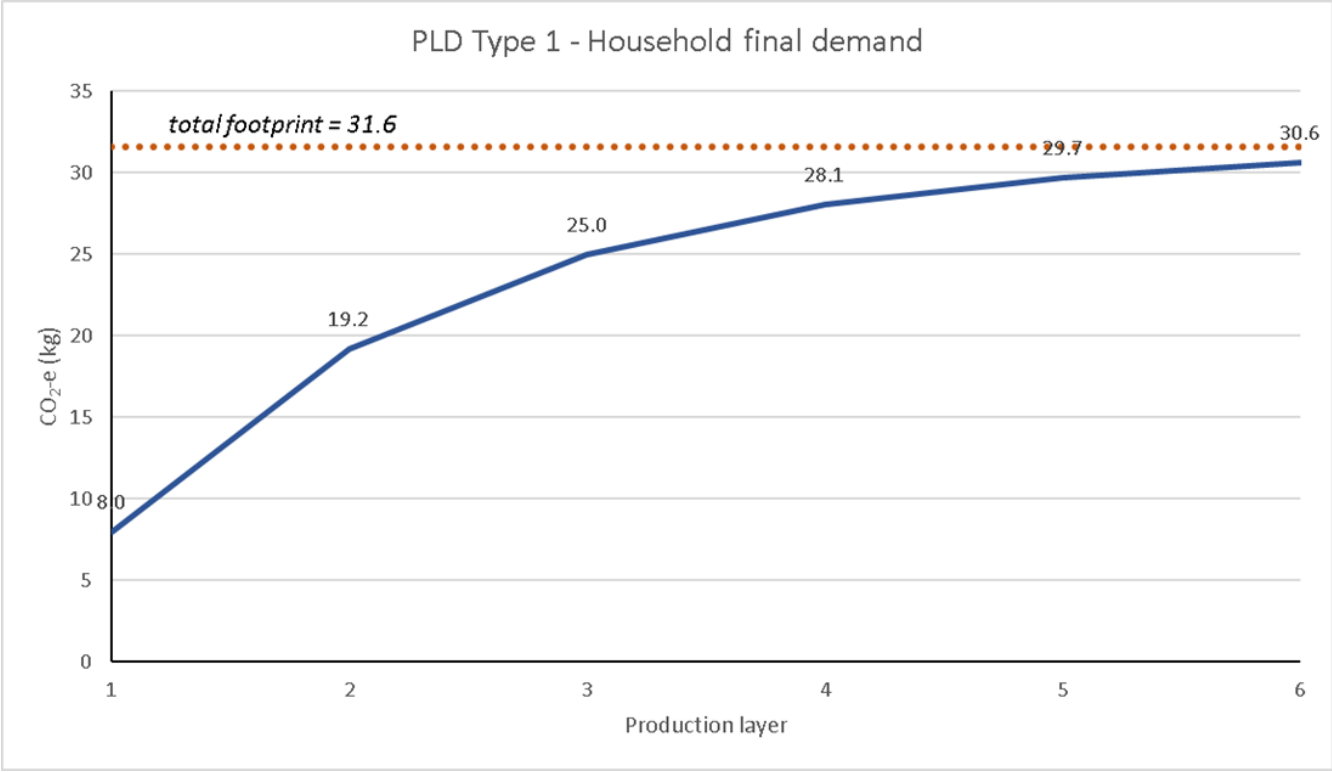

**Figure 1** Convergence of cumulative labor requirements for a A$1,000 expenditure on a new vehicle and on vehicle repairs. The initial ranking is reversed as the system boundary extends to include higher-order production layers. *Note:* emp-h = employment-hours.

# Decomposing the footprint calculation

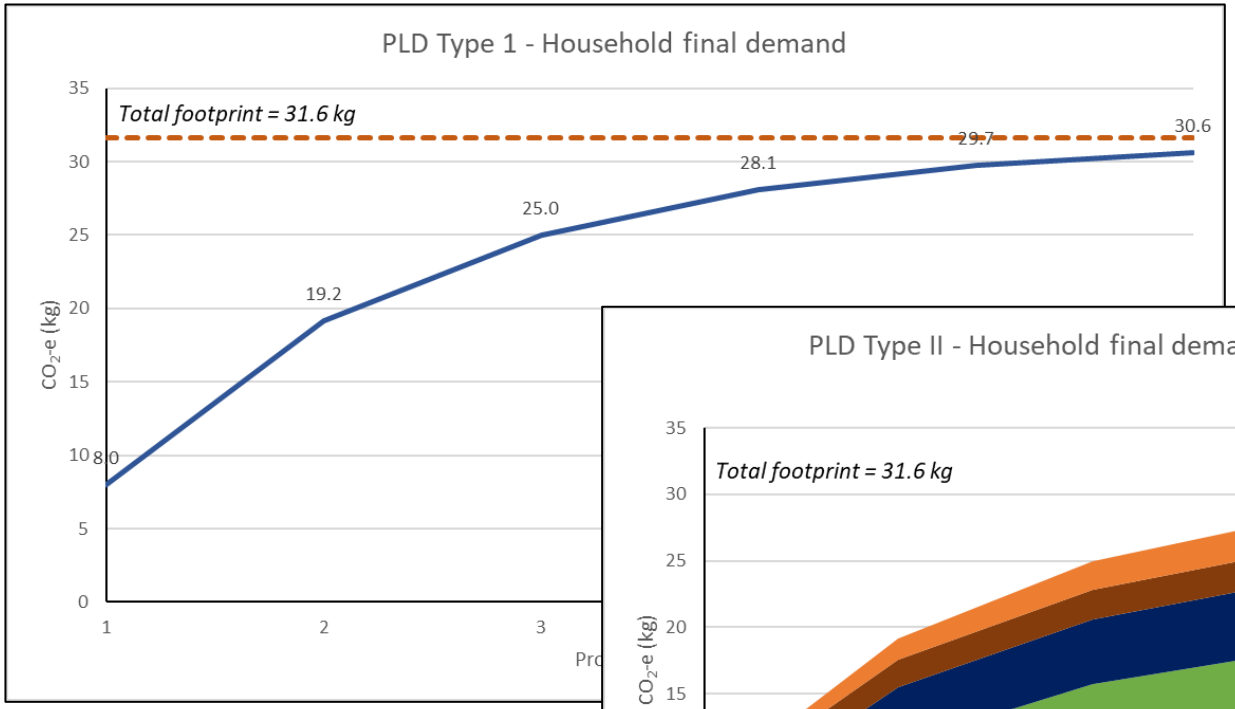
## Type I PLD

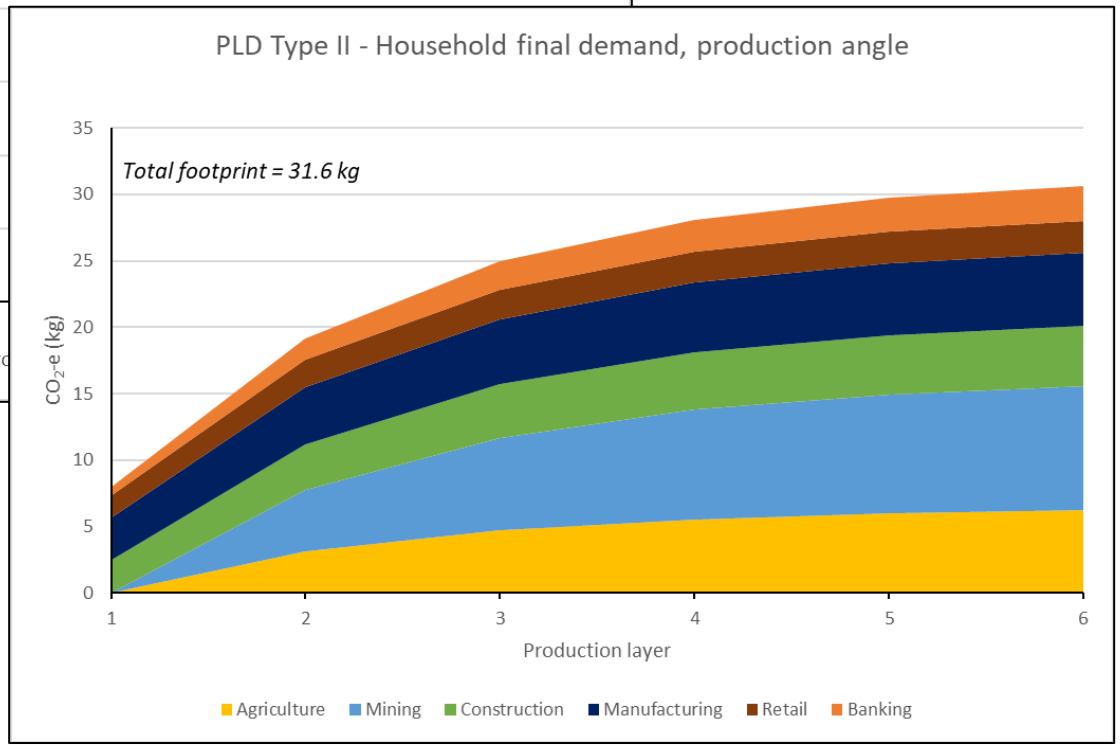
$$\mathbf{q}y + \mathbf{q}\mathbf{A}y + \mathbf{q}\mathbf{A}^2y + \mathbf{q}\mathbf{A}^3y + \mathbf{q}\mathbf{A}^4y + \cdots + \mathbf{q}\mathbf{A}^ny$$

Layer 1 | Layer 2 | Layer 3 | Layer 4 | Layer 5 | Layer *n*+1



# Two types of PLD decomposition



# RECAP - MATRIX ALGEBRA



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# Matrix operations

## Matrix multiplication

Two matrices can only by multiplied together using matrix multiplication when the number of columns in the first matrix is equal to the number of rows in the second matrix

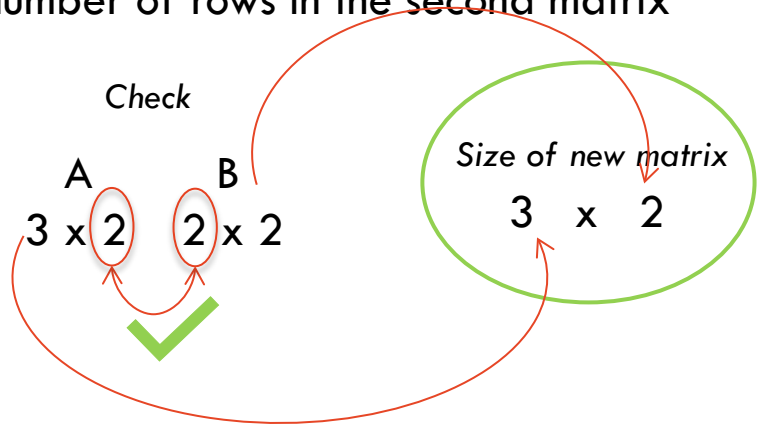
A

| | |
|---|---|
| 1 | 2 |
| 3 | 4 |
| 5 | 6 |

x

B

| | |
|---|---|
| 7 | 8 |
| 9 | 10 |



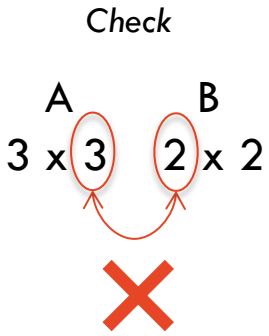

A

| | | |
|---|---|---|
| 1 | 2 | 7 |
| 3 | 4 | 8 |
| 5 | 6 | 9 |

x

B

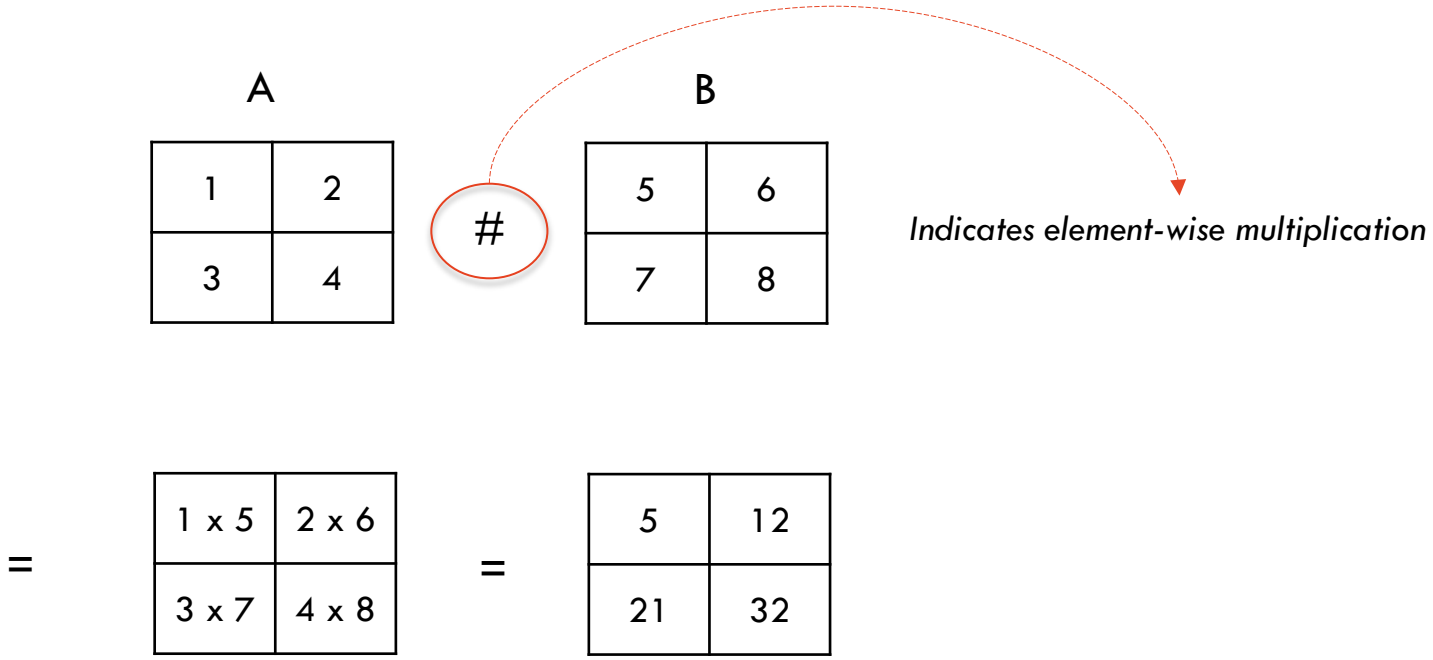
| | |
|---|---|
| 7 | 8 |
| 9 | 10 |



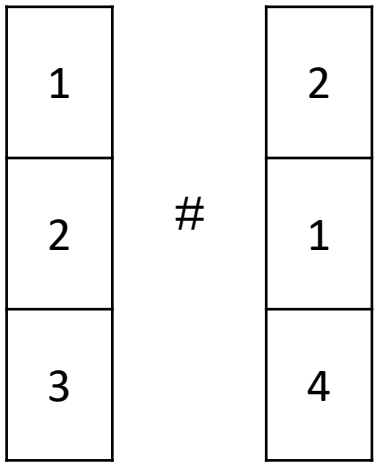
# Matrix operations

## Element-wise multiplication

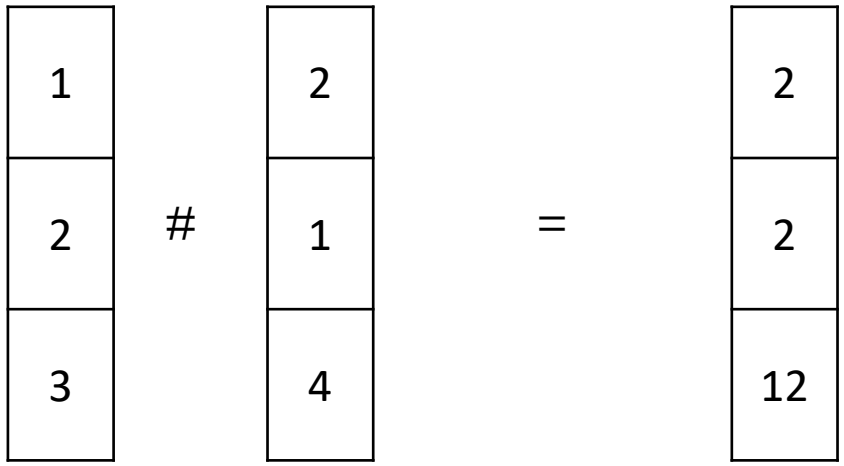
If two matrices are the same size, we can carry out element-wise multiplication



# Element-wise multiplication



# Element-wise multiplication



# Coming back to the PLD equation…

# Decomposing the footprint calculation

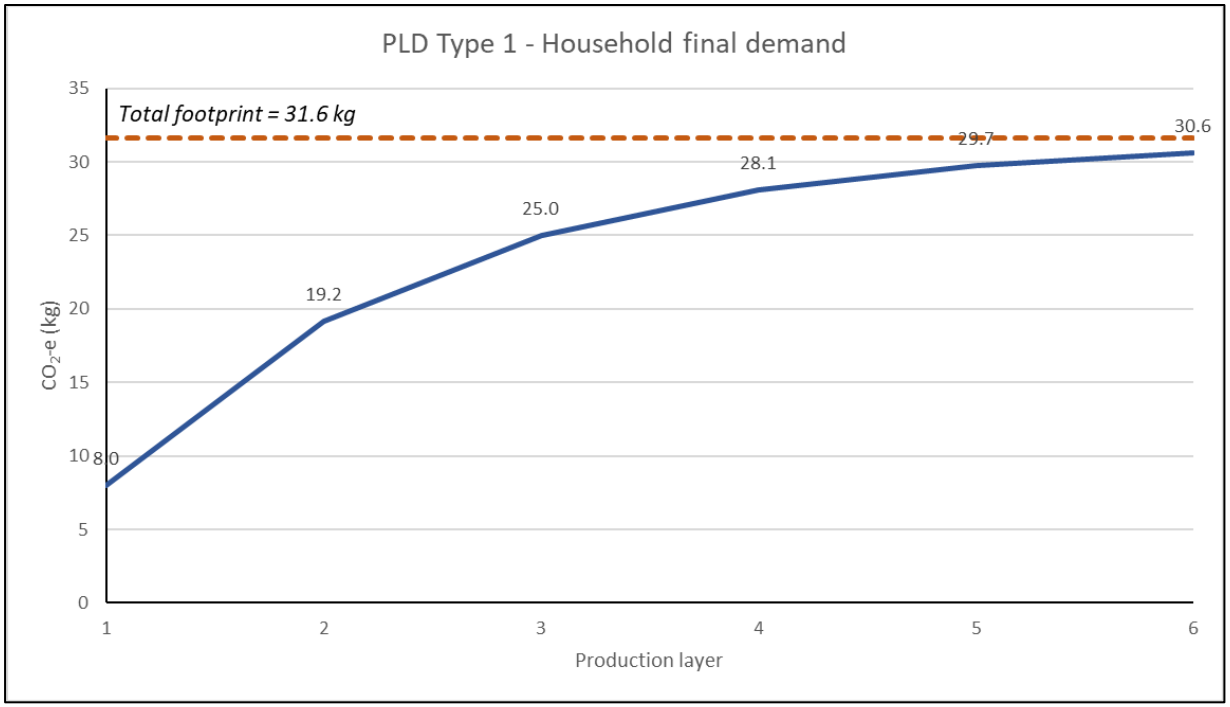
# Type I PLD

$$\underbrace{(\mathbf{q} * \mathbf{y})}_{(1x6)x(6x1)} + \underbrace{(\mathbf{q} * \mathbf{A} * \mathbf{y})}_{(1x6)x(6x6)x(6x1)} + \underbrace{(\mathbf{q} * \mathbf{A}^2 * \mathbf{y})}_{(1x6)x(6x6)x(6x1)} + \underbrace{(\mathbf{q} * \mathbf{A}^3 * \mathbf{y})}_{(1x6)x(6x6)x(6x1)} + \underbrace{(\mathbf{q} * \mathbf{A}^4 * \mathbf{y})}_{(1x6)x(6x6)x(6x1)} + \cdots + \underbrace{(\mathbf{q} * \mathbf{A}^n * \mathbf{y})}_{(1x6)x(6x6)x(6x1)}$$

$$\underbrace{\phantom{(\mathbf{q} * \mathbf{y})}}_{(1x1)} \quad \underbrace{\phantom{(\mathbf{q} * \mathbf{A} * \mathbf{y})}}_{(1x1)} \quad \underbrace{\phantom{(\mathbf{q} * \mathbf{A}^2 * \mathbf{y})}}_{(1x1)} \quad \underbrace{\phantom{(\mathbf{q} * \mathbf{A}^3 * \mathbf{y})}}_{(1x1)} \quad \underbrace{\phantom{(\mathbf{q} * \mathbf{A}^4 * \mathbf{y})}}_{(1x1)} \quad \underbrace{\phantom{(\mathbf{q} * \mathbf{A}^n * \mathbf{y})}}_{(1x1)}$$

Layer 1 | Layer 2 | Layer 3 | Layer 4 | Layer 5 | Layer *n*+1

* mmult (matrix multiplication)



# Decomposing the footprint calculation
# Type II PLD

Using matrix transposition and element-wise multiplication

$\boldsymbol{q} =$

| 0.4 | 0.3 | 0.45 | 0.66 | 0.75 | 0.25 |
|---|---|---|---|---|---|

(1 x 6)

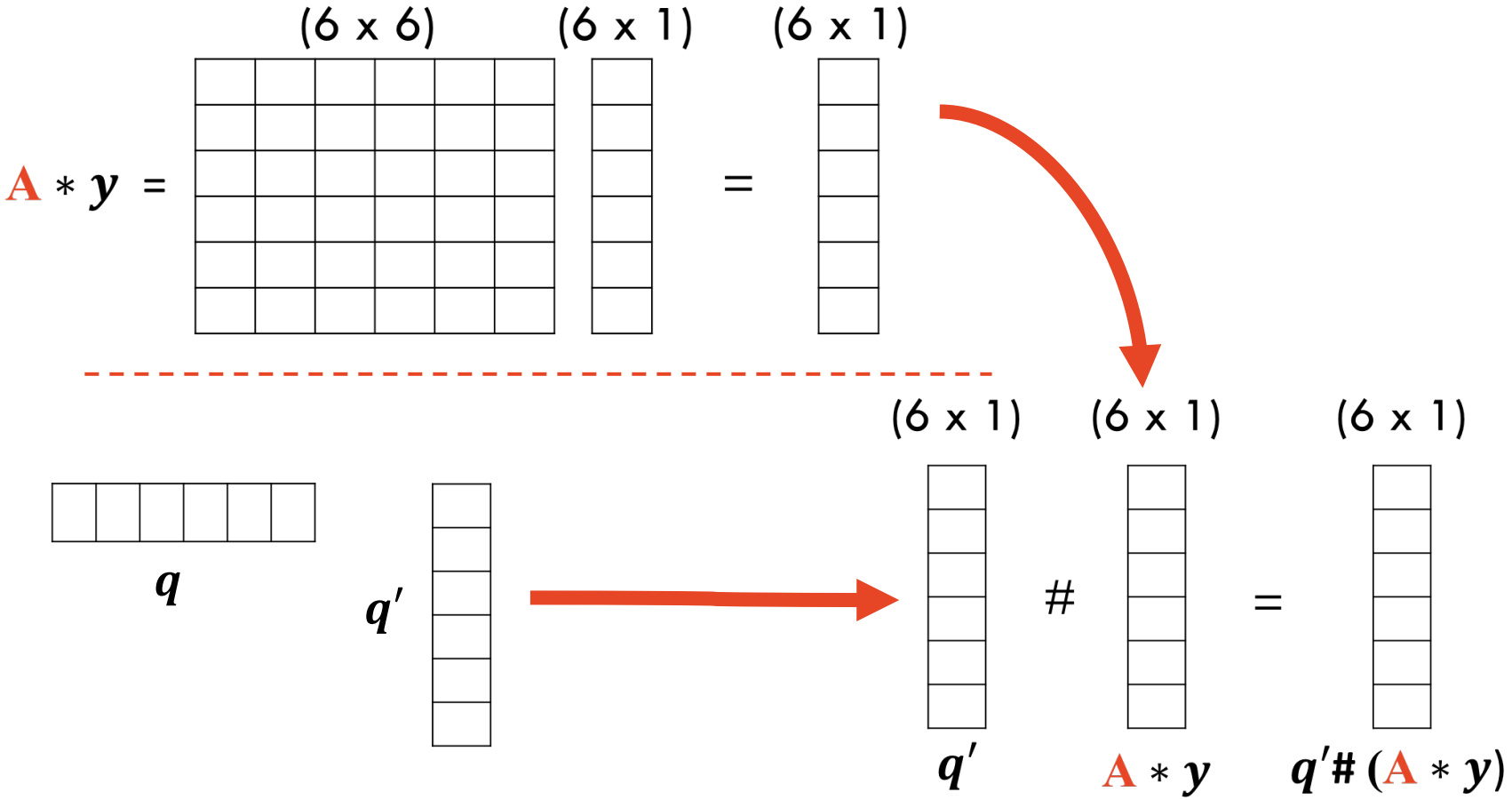
$\boldsymbol{q}' =$

| 0.4 |
|---|
| 0.3 |
| 0.45 |
| 0.66 |
| 0.75 |
| 0.25 |

(6 x 1)

# Decomposing the footprint calculation
# Type II PLD

Using matrix transposition and element-wise multiplication



# Decomposing the footprint calculation
# Type II PLD

Using matrix transposition and element-wise multiplication

$$(\mathbf{q}'\#\boldsymbol{y})+\left(\mathbf{q}'\#\mathbf{A}*\boldsymbol{y}\right)+(\mathbf{q}'\#\mathbf{A}^2*\boldsymbol{y})+\left(\mathbf{q}'\#\mathbf{A}^3*\boldsymbol{y}\right)+(\mathbf{q}'\#\mathbf{A}^4*\boldsymbol{y})+\cdots+(\mathbf{q}'\#\mathbf{A}^n*\boldsymbol{y})$$

$*$ matrix multiplication

$.*$ or $\otimes$ or $\#$ element wise multiplication

$'$ transpose

Use this version to avoid confusion between * and .*

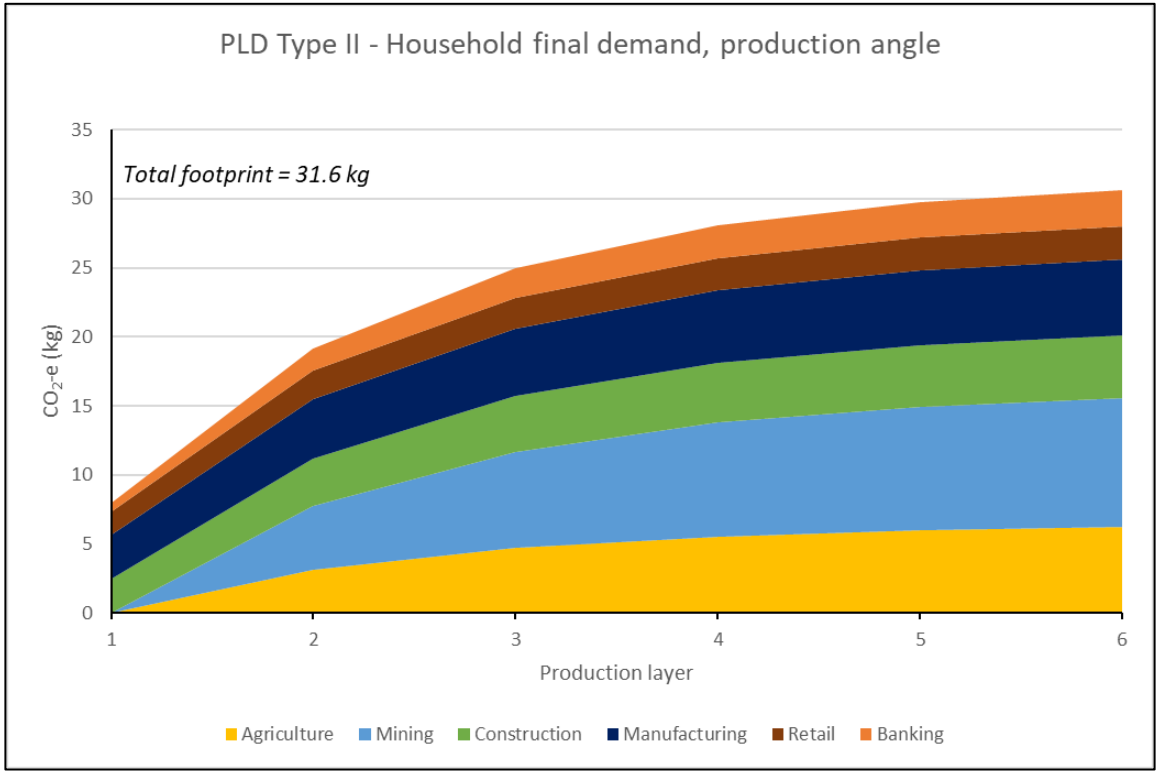
# Decomposing the footprint calculation
# Type II PLD

(6x1)x(6x1) (6x1)#[(6x6)x(6x1)] (6x1)#[(6x6)x(6x1)] (6x1)#[(6x6)x(6x1)] (6x1)#[(6x6)x(6x1)] (6x1)#[(6x6)x(6x1)]

$$(\mathbf{q}'\#\boldsymbol{y})+(\mathbf{q}'\#\mathbf{A}*\boldsymbol{y})+(\mathbf{q}'\#\mathbf{A}^2*\boldsymbol{y})+(\mathbf{q}'\#\mathbf{A}^3*\boldsymbol{y})+(\mathbf{q}'\#\mathbf{A}^4*\boldsymbol{y})+\cdots+(\mathbf{q}'\#\mathbf{A}^n*\boldsymbol{y})$$

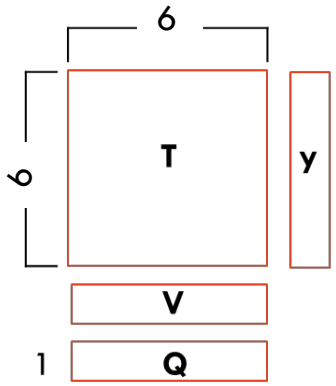
(6x1) (6x1) (6x1) (6x1) (6x1) (6x1)

Layer 1 Layer 2 Layer 3 Layer *4* Layer *5* Layer *n+1*



# How do the dimensions work in PLD I and PLD II?

# PLD Type I




## Layer 1

$$q * y$$

Size check:

(1 x 6) * (6 x 1) => (1 x 1)

# PLD Type II

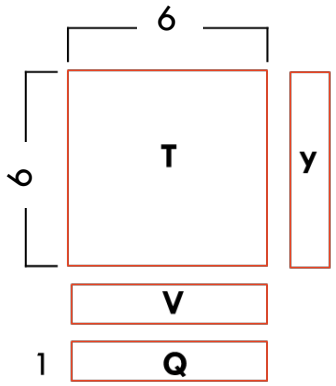
## Layer 1

$$q' \# y$$

Size check:

(6 x 1) # (6 x 1) => (6 x 1)

## PLD Type I




## PLD Type II

### Layer 1

$$q * y$$

Size check:

(1 x 6) * (6 x 1) => (1 x 1)

### Layer 1

$$q' \# y$$

Size check:

(6 x 1) # (6 x 1) => (6 x 1)

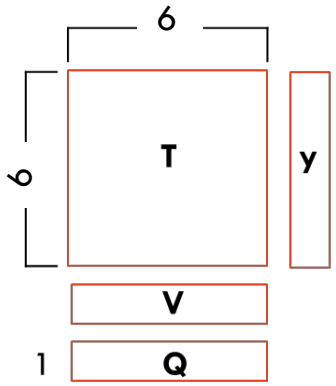
### Layer 2

$$q * A * y$$

### Layer 2

$$q' \# A * y$$

## PLD Type I




### Layer 1

$$q * y$$

Size check:

(1 x 6) * (6 x 1) => (1 x 1)

### Layer 2

$$q * \boxed{A * y}$$

Size check 1:

(6 x 6) * (6 x 1) => (6 x 1)

## PLD Type II

### Layer 1

$$q' \# y$$

Size check:

(6 x 1) # (6 x 1) => (6 x 1)

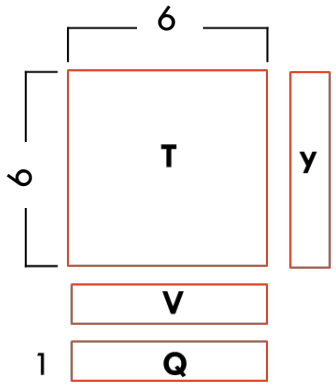
### Layer 2

$$q' \# \boxed{A * y}$$

Size check 1:

(6 x 6) * (6 x 1) => (6 x 1)

# PLD Type I




## Layer 1

$$q * y$$

Size check:

(1 x 6) * (6 x 1) => (1 x 1)

## Layer 2

$$q * A * y$$

Size check 2:

(1 x 6) * (6 X 1) => (1 x 1)

# PLD Type II

## Layer 1

$$q' \# y$$

Size check:

(6 x 1) # (6 x 1) => (6 x 1)

## Layer 2

$$q' \# A * y$$

Size check 2:

(6 x 1) # (6 x 1) => (6 x 1)

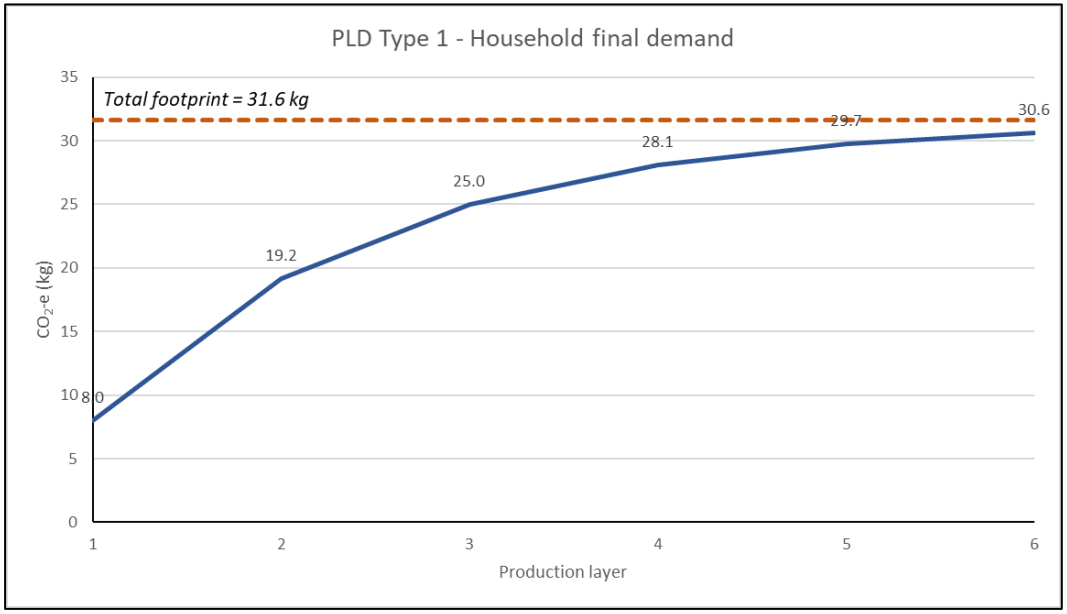
matrices are
the same size

# Decomposing the footprint calculation
# Type I PLD

$$(\mathbf{q} * \boldsymbol{y}) + (\mathbf{q} * \mathbf{A} * \boldsymbol{y}) + (\mathbf{q} * \mathbf{A}^2 * \boldsymbol{y}) + (\mathbf{q} * \mathbf{A}^3 * \boldsymbol{y}) + (\mathbf{q} * \mathbf{A}^4 * \boldsymbol{y}) + \cdots + (\mathbf{q} * \mathbf{A}^n * \boldsymbol{y})$$

Layer 1 | Layer 2 | Layer 3 | Layer 4 | Layer 5 | Layer $n + 1$

* mmult (matrix multiplication)

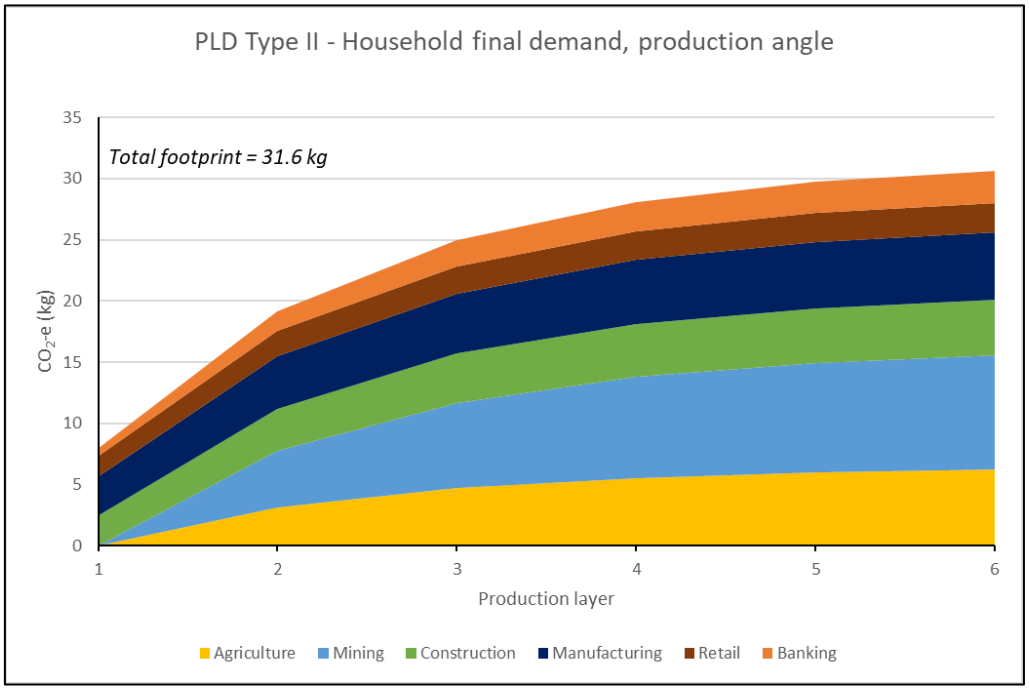


# Decomposing the footprint calculation
# Type II PLD

$$(\mathbf{q}'\#\boldsymbol{y})+\left(\mathbf{q}'\#\mathbf{A}*\boldsymbol{y}\right)+(\mathbf{q}'\#\mathbf{A}^2*\boldsymbol{y})+\left(\mathbf{q}'\#\mathbf{A}^3*\boldsymbol{y}\right)+(\mathbf{q}'\#\mathbf{A}^4*\boldsymbol{y})+\cdots+(\mathbf{q}'\#\mathbf{A}^n*\boldsymbol{y})$$

Layer 1 | Layer 2 | Layer 3 | Layer 4 | Layer 5 | Layer $n+1$

\# element-wise multiplication



# Using PLD to calculate Scope 1 & Scope 2 emissions

Calculate using physical units and emissions factors, or . . .

## Calculating Scope 1

Scope 1 emissions can be calculated using emissions factors and information on the amount of fuel used.

Note the units carefully – we need to do two conversions here:

1. Volume -> GJ of energy
2. GJ of energy -> $CO_2$-e

*Table 9 Direct (scope 1) and indirect (scope 3) emission factors for the consumption of transport fuels in different transport equipment*

| Transport type | Fuel combusted | Energy Content Factor (GJ per unit of fuel) | Scope 1 Emission Factor (kg $CO_2$-e/GJ) $CO_2$ | Scope 1 Emission Factor (kg $CO_2$-e/GJ) $CH_4$ | Scope 1 Emission Factor (kg $CO_2$-e/GJ) $N_2O$ | Scope 1 Emission Factor (kg $CO_2$-e/GJ) Combined gases | Scope 3 Emission Factor (kg $CO_2$-e/GJ) |
|---|---|---|---|---|---|---|---|
| Cars and light commercial vehicles | Gasoline | 34.2 | 67.4 | | | | |
| Cars and light commercial vehicles | Diesel oil | 38.6 | 69.9 | | | | |
| Cars and light commercial vehicles | Liquefied petroleum gas (LPG) | 26.2 | 60.2 | | | | |
| Cars and light commercial vehicles | Fuel oil | 39.7 | 73.6 | | | | |
| Cars and light commercial vehicles | Ethanol | 23.4 | 0.0 | | | | |
| Cars and light commercial vehicles | Biodiesel | 34.6 | 0.0 | | | | |
| Cars and light commercial vehicles | Renewable diesel | 38.6 | 0.0 | | | | |

https://www.dcceew.gov.au/climate-change/publications/national-greenhous

## Calculating Scope 2

Scope 2 emissions can be calculated using emissions factors and information on kWh of electricity purchased.

*Table 1 Indirect (scope 2 and scope 3) location-based emission factors associated with the consumption and losses of purchased or acquired electricity*

| State, Territory or grid description | Scope 2 Emission Factors (kg $CO_2$-e/kWh) | Scope 3 Emission Factors (kg $CO_2$-e/kWh) |
|---|---|---|
| New South Wales and Australian Capital Territory | 0.60 | 0.07 |
| Victoria | 0.74 | 0.11 |
| Queensland | 0.65 | 0.11 |
| South Australia | 0.21 | 0.03 |
| Western Australia - South West Interconnected System (SWIS) | 0.45 | 0.10 |
| Western Australia - North Western Interconnected System (NWIS) | 0.56 | 0.09 |
| Tasmania | 0.23 | 0.03 |
| Northern territory - Darwin Katherine Interconnected System (DKIS) | 0.55 | 0.13 |
| National | 0.59 | 0.10 |

**Sources:** Primary data sources comprise National Greenhouse and Energy Reporting (Measurement) Determination 2008 (Schedule 1), Australian Energy Statistics, Clean Energy Regulator, and AEMO data and Department of Climate Change, Energy, the Environment and Water.

https://www.dcceew.gov.au/climate-change/publications/national-greenhouse-accounts-factors-2026

# Using PLD to calculate Scope 2 emissions

$(\mathbf{q} * \boldsymbol{y})$

Layer 1

Impact due to the direct consumption being analysed

$$\text{Scope 2} = (\mathbf{q}_{elect} * \mathbf{y}_{elect})$$

These are estimations based on sector averages and will not be as accurate as the activity-based method described in week 8, but in the absence of physical data can provide a good initial estimate.

# Adding in Scope 1 emissions

$$\text{Scope 1} = \mathbf{q}_{\text{my_sector}} * \mathbf{x}_{\text{me}}$$



Impact due to direct operations per monetary unit of total output

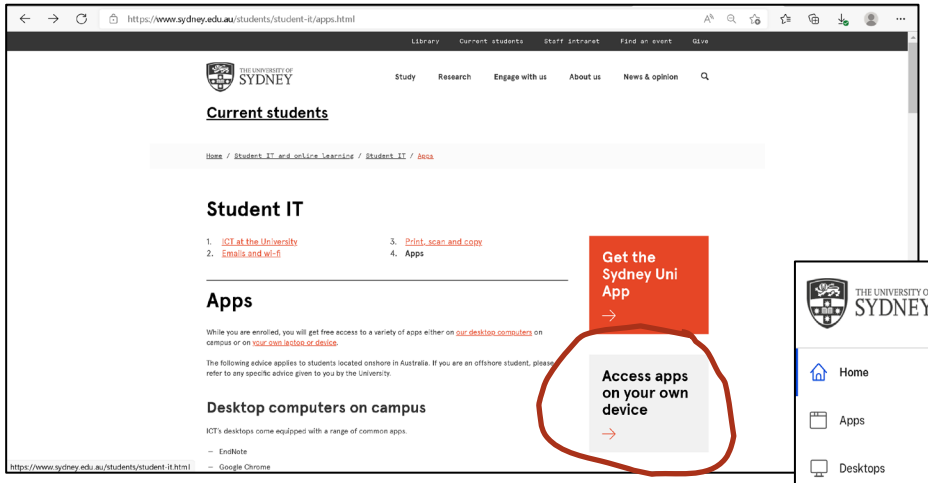
My revenue, also known as total output

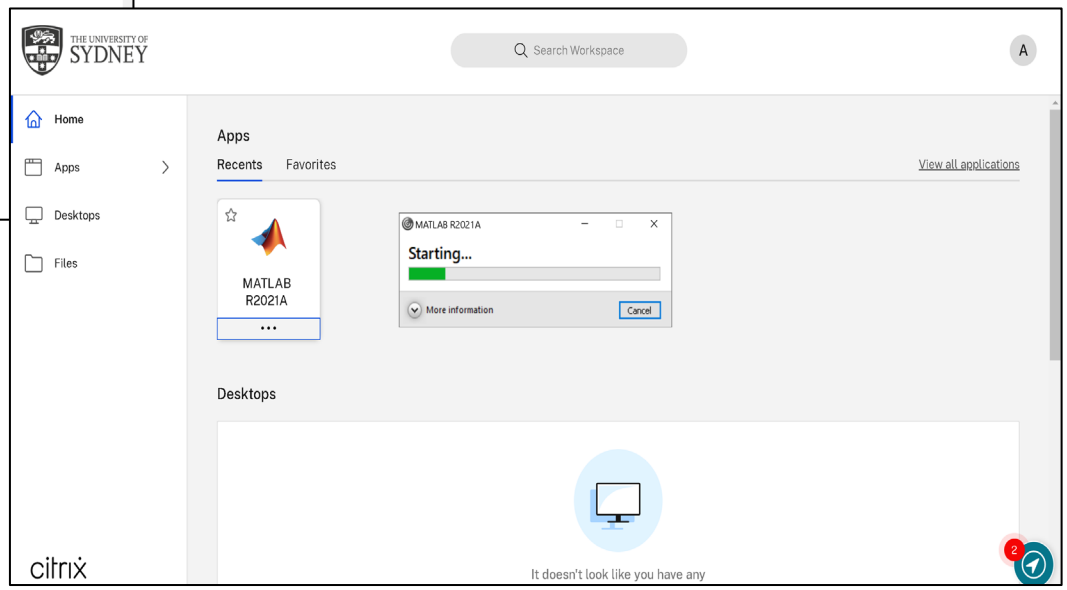
These are estimations based on sector averages and will not be as accurate as the activity-based method described in week 8, but in the absence of physical data can provide a good initial estimate.

# Getting started with MATLAB

https://www.sydney.edu.au/students/student-it/apps.html

Sign in with your Unikey



# Getting started with MATLAB

https://www.sydney.edu.au/students/student-it/apps.html

Sign in with your Unikey

## On your own device

### Apps to download

We provide a range of apps that can be downloaded for your personal device. Many can be downloaded and used offline.

LinkedIn Learning +

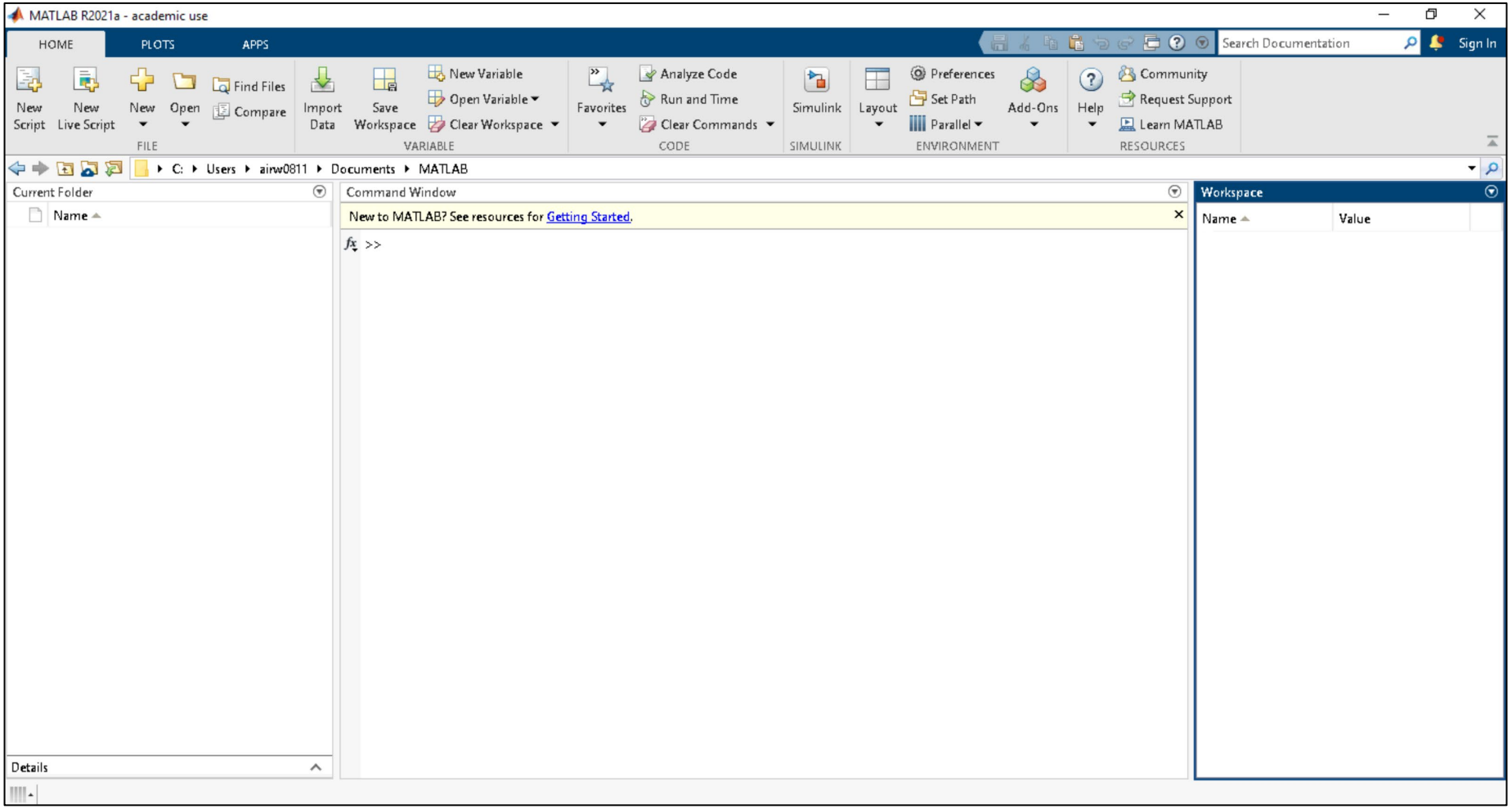
MATLAB −

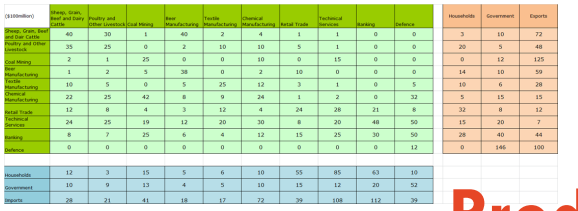
MATLAB helps you with numerical computation, visualisation and programming. With MATLAB you can analyse data, develop algorithms and create models.

MATLAB is available on Windows and Mac OS X.

Product information | Download MATLAB

# Getting started with MATLAB





| ($10Million) | Crops, Grain, Beef and Dairy Cattle | Poultry and Other Livestock | Coal Mining | Iron Manufacturing | Textile Manufacturing | Chemical Manufacturing | Retail Trade | Technical Services | Banking | Defence |
|---|---|---|---|---|---|---|---|---|---|---|
| Crops, Grain, Beef and Dair Cattle | 40 | 30 | 1 | 40 | 2 | 4 | 1 | 1 | 0 | 0 |
| Poultry and Other Livestock | 35 | 25 | 0 | 2 | 10 | 10 | 5 | 1 | 0 | 0 |
| Coal Mining | 2 | 1 | 25 | 0 | 0 | 10 | 0 | 15 | 0 | 0 |
| Iron Manufacturing | 1 | 2 | 5 | 38 | 0 | 2 | 10 | 0 | 0 | 0 |
| Textile Manufacturing | 10 | 5 | 0 | 5 | 25 | 12 | 3 | 1 | 0 | 5 |
| Chemical Manufacturing | 22 | 25 | 42 | 8 | 9 | 24 | 1 | 2 | 0 | 32 |
| Retail Trade | 12 | 8 | 4 | 3 | 12 | 4 | 24 | 28 | 21 | 8 |
| Technical Services | 24 | 25 | 19 | 12 | 20 | 30 | 8 | 20 | 48 | 50 |
| Banking | 8 | 7 | 25 | 6 | 4 | 12 | 15 | 25 | 30 | 50 |
| Defence | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 12 |
| Households | 12 | 3 | 15 | 5 | 6 | 10 | 55 | 85 | 63 | 10 |
| Government | 10 | 9 | 13 | 4 | 5 | 10 | 15 | 12 | 20 | 52 |
| Imports | 28 | 21 | 41 | 18 | 17 | 72 | 39 | 108 | 112 | 39 |

| Households | Government | Exports |
|---|---|---|
| 3 | 10 | 72 |
| 20 | 5 | 48 |
| 0 | 12 | 125 |
| 14 | 10 | 59 |
| 10 | 6 | 28 |
| 5 | 15 | 15 |
| 32 | 8 | 12 |
| 15 | 20 | 7 |
| 28 | 40 | 44 |
| 0 | 146 | 100 |

# Tutorial Activity
# Production Layer Decomposition

## Use the input-output table from weeks 5, 6 & 8 to complete the following:

1. **Choose one final demand vector**
   Calculate the footprint for that final demand vector using f = **qLy**

2. **For the first 7 production layers, calculate A^i, where i = 1,2,..6.**
   See if the numbers in the subsequent A^i matrices eventually become smaller. The last one (A^6) should very small compared to the first one.

3. **Calculate your 7 production layer footprints f^i, where i=1,2,..,6. Use the different terms of the formula f=qy+qAy+qA^2y+... for this.**
   You should have seven different production layer footprint vectors.

4. **Add all of these 7 footprint vectors and compare the sum with your result from step 1) above.**
   They should be almost equal , but which one is bigger?

5. **Choose one of the indicators from the Q block, and generate a plot showing the cumulative footprints for each production layer.** For production layer 1, show footprint 1. For production layer 2, show the sum of footprints 1+2 and so on.

# Homework for Week 9

## Homework week 9 (Due: October 28, 2026)

**The aim of this exercise is to observe that production layer decomposition does actually split the footprint into several parts. Each of these parts refer to the footprint in one of the production layers. Use your 14-sector table to complete the following (Note: this is TYPE 1 PLD and you are required to choose one indicator):**

1. **Preparation:**

Choose a final demand vector of your choice. This can be one of the standard ones (Households, Government or Exports), or a customised one (i.e. your shopping basket).

2. **Calculate the footprint for one of your indicators for that final demand vector (using f = qLy)**
3. **Calculate A^i, i=1,...,6**

See if the numbers in the subsequent A^i matrices eventually become smaller. The last one (A^6) should very small compared to the first one.

4. **Calculate 7 production layer footprints for that indicator using the same final demand vector f^i, i=1,...,7. Use the different terms of the formula f=qy+qAy+qA^2y+... for this.**

Now you should have seven different production layer footprint vectors for your indicator.

5. **Add all of these 7 footprint vectors and compare the sum with your result from step 2) above. They should be almost equal. But which one is bigger?**
6. **Generate a plot showing the cumulative footprints for each production layer for your indicator.**

For production layer 1, show footprint 1. For production layer 2, show the sum of footprints 1+2 and so on.

**EXPLANATION:** Explain the concept of production layer decomposition in words (100 words). Why does your graph converge after a number of production layers? (100 words)

# PHYS5033 organisation – assessment tasks

## Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

# Key takeaways

- The footprint equation can be decomposed using the Taylor series expansion

$$(\mathbf{I}-\mathbf{A})^{-1} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 + \cdots + \mathbf{A}^n$$

- This enables us to calculate the footprint at each production layer

- We can deconstruct this further by sector using transposition and element-wise multiplication

More to come:

- Industry guest lecture
- Structural Path Analysis